+ 1 207 880 8545
sebf233@gmail.com

Sebastian H.L. Francis

101 Delano Park
Cape Elizabeth, ME 04107

EDUCATION

2025 - 2029 **University of California, Berkeley** — Berkeley, CA
Bachelor of Arts in Applied Math, Minor in Computer Science

EXPERIENCE

2022 - 2025 **Open Bench Project STEM Camps** — Portland, ME
Lead Counselor
- Led week-long STEM camps for 100+ students ages 8-13, developing curriculum to teach engineering and design skills using OnShape, Fusion 360, Autodesk inventor, 3D printing, CO2 laser cutting, MIG welding, CNC, and Arduino microcontrollers

2024 - 2025 **Inspire Girls Coding Bootcamp Coach: Zambia and Nigeria** — Remote
- Created instructional videos introducing the basics of programming in Python to 60+ underprivileged school girls in Zambia and Nigeria

2021 - 2025 **World Computer Exchange** — Cape Elizabeth, ME
- Founded Maine's first World Computer Exchange chapter, creating a community initiative that sourced and refurbished computers for 100+ underprivileged schools, earning a $1000 grant.

2025 - Present **Crosscheck** — Remote
- Consulting for Crosscheck, digital checklists for pilots

PROJECTS

2024 **3D-Printed Hydrogen Internal Combustion Engine** — Bangor, ME
- Designed and printed hydrogen internal combustion engine using Onshape, Fusion 360, and Arduino, earning the Yale Science & Engineering Association Award at the 2024 Maine State Science Fair

2021 - 2023 **Liquid Oxygen Rocket** — Cape Elizabeth, ME
- Built 12-foot liquid oxygen rocket featuring programmable sensors, custom ignition and remote fueling systems
- Led development of sensor data collection systems using Adafruit sensors, a Raspberry Pi microcontroller, and Python
- Invited to by local biofuel company to present rocket

2019 **NASA JPL Mars Rover Replica** — Portland, ME
- Built a replica Mars Rover using the NASA JPL specifications

LEADERSHIP

2025 - **Aviation Club of Berkeley** — Berkeley, CA
- Helped prepare club members for the FAA commercial airplane checkride

SKILLS

Technical	Python, Java, Firebase, HTML, CSS, JavaScript, Autodesk Inventor, Onshape, Autodesk Fusion 360, Blender, Unreal Engine, CO2 laser cutter, CNC, MIG welding, 3D printers, Arduino, Raspberry Pi
Aviation	FAA Fixed Wing and Rotorcraft Private Pilot Certificate, Tailwheel Endorsement, Glider Student